AIM:K-Nearest Neighbors (K-NN)

- Implement the K-NN algorithm for classification or regression.
- Apply the K-NN algorithm to a given dataset and predict the class or value
- for test data.
- Evaluate the accuracy or error of the predictions and analyze the results.

CODE:

```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler


df = pd.read_csv("C:/Users/hrane/Downloads/online food delivery_107.csv")
df.columns = df.columns.str.strip()
df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)


df["Output"] = df["Output"].map({"Yes": 1, "No": 0})


X = df.drop(["Output", "Feedback"], axis=1, errors="ignore")
y = df["Output"]


X = pd.get_dummies(X, drop_first=True)


scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y
)


k_values = range(1, 16)
accuracies = []

for k in k_values:
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_train, y_train)
    predictions = knn.predict(X_test)
    accuracies.append(accuracy_score(y_test, predictions) * 100)
```

RIYA T107

```
optimal_k = 5
knn_final = KNeighborsClassifier(n_neighbors=optimal_k)
knn_final.fit(X_train, y_train)
y_pred = knn_final.predict(X_test)


print("=" * 60)
print(f"--- K-NN Classification Analysis (Tested at K={optimal_k}) ---")
print(f"Final Model Test Accuracy: {accuracy_score(y_test, y_pred) * 100:.2f}%")
print("=" * 60)
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=["No", "Yes"]))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))


plt.figure(figsize=(10, 5))
plt.plot(
    k_values,
    accuracies,
    color="blue",
    linestyle="dashed",
    marker="o",
    markerfacecolor="red",
    markersize=8,
)


plt.xlabel("Number of Neighbors (K)")
plt.ylabel("Testing Accuracy (%)")
plt.xticks(k_values)
plt.grid(True, linestyle=":", alpha=0.6)
plt.show()
```

OUTPUT:

RIYA T107

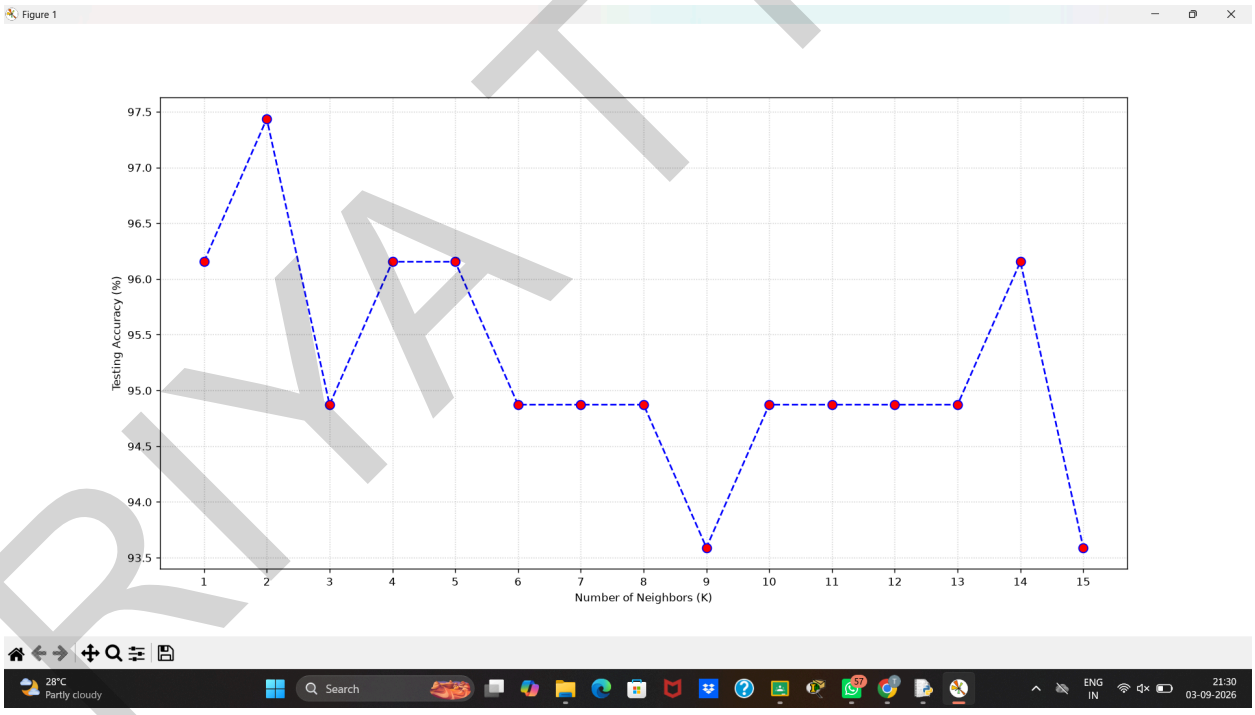
```
========== RESTART: C:\Users\hrane\Documents\AI_K.p
============================================================
--- K-NN Classification Analysis (Tested at K=5) ---
Final Model Test Accuracy: 96.15%
============================================================

Classification Report:
              precision    recall  f1-score   support

          No       0.94      0.88      0.91        17
         Yes       0.97      0.98      0.98        61

    accuracy                           0.96        78
   macro avg       0.95      0.93      0.94        78
weighted avg       0.96      0.96      0.96        78


Confusion Matrix:
[[15  2]
 [ 1 60]]
```



RIYA T107